Team Problems

1. Let S be the set of positive factors of 2010. Compute the median of S.

2. Point P is located in the interior of rectangle $ABCD$ and $PA = 56$, $PB = 25$ and $PC = 33$. Compute PD.

3. At a meeting of the Nuclear Powerplant Workers of America, every person has 3, 4, 5, 6, or 7 fingers on each hand, with the probability of having k fingers on a hand being $\frac{2^{2-|5-k|}}{10}$. If the number of fingers on a worker's left hand is independent of the number of fingers on the worker's right hand, compute the probability that a NPWA member has at least 10 fingers.

4. Given an unlimited set of identical 1×1, 1×2, and 1×3 tiles, compute the number of ways to cover a 1×8 rectangle with the tiles without overlapping tiles and with all tiles entirely inside the rectangle. Rotations of a set of tiles are considered distinct (for examples, two 1×3 tiles followed by a 1×2 tile is a distinct tiling from a 1×2 tile followed by two 1×3 tiles).

5. For two positive integers x and y, $x^2 - y^2 = 240$. Compute the (positive) difference between the largest and the smallest possible values of xy.

6. Walls are erected on a frictionless surface in the form of an equilateral triangle $\triangle ABC$ of side length 1. A tiny super bouncy ball is fired from vertex A at side $\overline{BC}$. The ball bounces off of 9 walls before hitting a vertex of the triangle for the first time. Compute the total distance travelled by the ball upon arrival.

7. A hexagonal pyramid has a regular hexagon of side length 1 for a base, and its apex is located a distance 1 directly above the center of the hexagon. A sphere inscribed inside the pyramid touches every face of the pyramid. Compute the radius of the inscribed sphere.

8. The function $f(x) = \dfrac{x^4 + 3x^3 - 4x^2}{x^2 + 4x - 4}$ is negative over two different intervals of the real numbers. Compute the sum of the lengths of the two intervals.

9. The set of points $S = \{(x, y) : |x| + |x - y| + |x + y| \leq k\}$ has area 10 in the xy-plane. Compute k.

10.
$$\begin{array}{cccc} & A & B & C \\ & \times & D & E \\ & F & E & C \\ D & E & C & \\ \hline 4 & 6 & B & C \end{array}$$

In the multiplication problem above, A, B, C, D, E, and F are distinct, non-zero digits. Compute the six-digit number $\underline{A}\,\underline{B}\,\underline{C}\,\underline{D}\,\underline{E}\,\underline{F}$.